

D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Subject Name: Database Management System

Week 5

Module 3 :NORMALIZATION

Faculty Name : DR. UTTAM WAGHMODE

Index -

Lecture 28 –NORMALIZATION

Lecture 29 –1NF,2NF

Lecture 30 – 3NF,BCNF



D Y PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

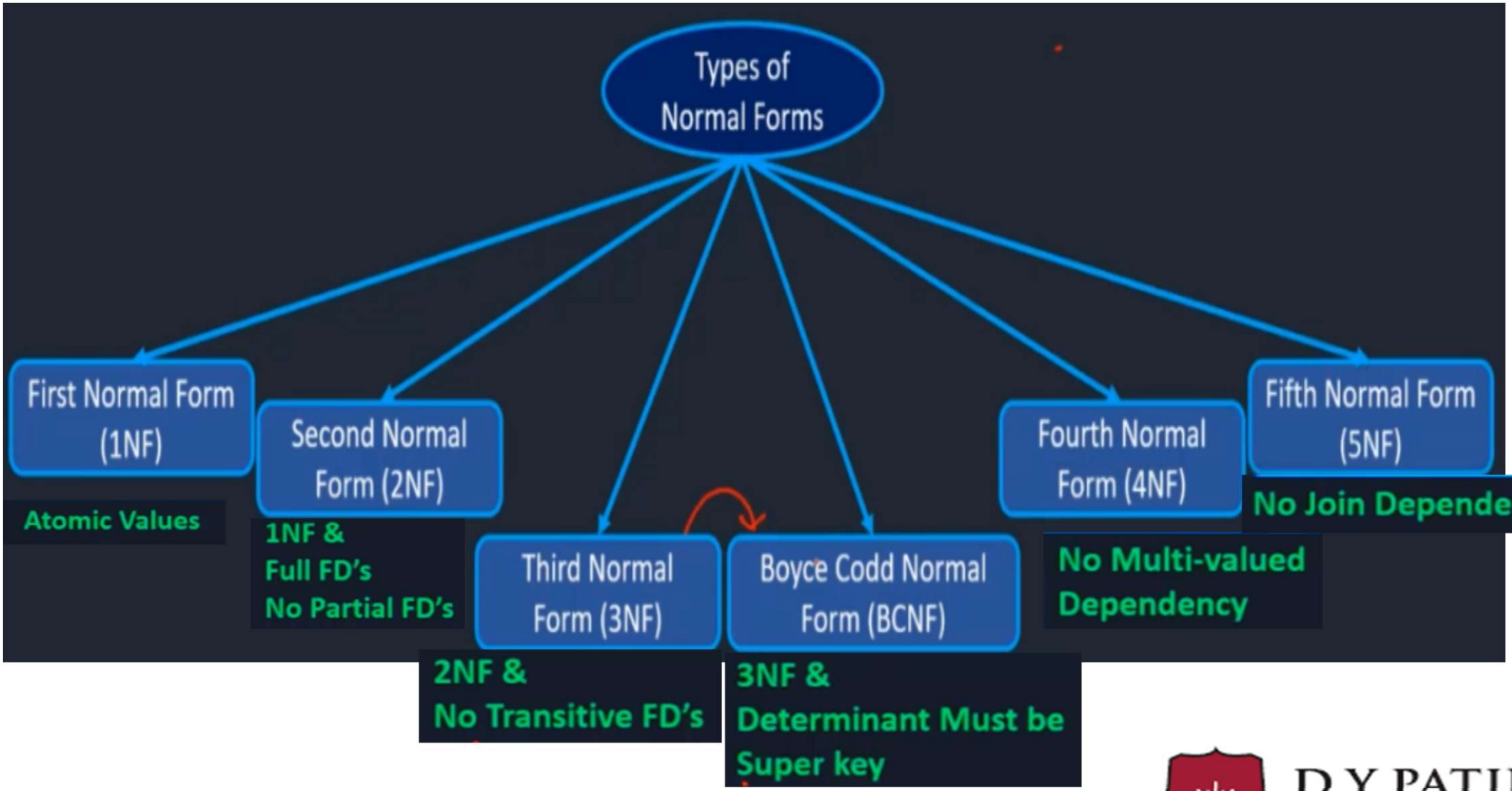
Lecture No: 28: NORMALIZATION



NORMALIZATION

- It is a process of organizing the data in the database.
- It is used to reduce the redundancy from a relation or set of relations
- It is used to eliminate the anomalies (insertion, deletion and update anomalies)
- It divides the larger relation into smaller relation without missing/duplicating the data and link them accordingly.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI



D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

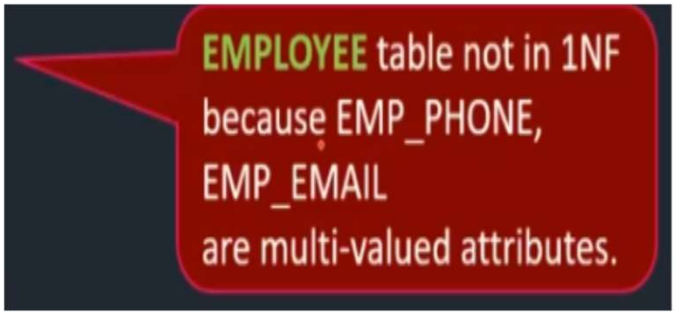
Lecture No: 29: 1NF & 2NF NORMALIZATION



1NF

- A relation in a 1NF if it contains atomic value (single valued attributes). No multiple values.
- It will not allow multivalued attribute and composite attributes , and their combination

EMP_NO	EMP_NAME	EMP_PHONE	EMP_EMAIL
50	Ravi	5012345678, 5023456789	abc@gmail.com
60	John	6012345678, 6023456789	xyz@gmail.com, pqr@gmail.com



D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

EMP_NO	EMP_NAME	EMP_PHONE	EMP_EMAIL
50	Ravi	5012345678	abc@gmail.com
50	Ravi	5023456789	abc@gmail.com
60	John	6012345678	xyz@gmail.com,
60	John	6012345678	pqr@gmail.com
60	John	6023456789	xyz@gmail.com,
60	John	6023456789	pqr@gmail.com

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

2NF

A relation is in 2NF, then it is in

- First Normal Form and
- Every non-primary-key attribute is fully functionally dependent on the primary key.

OR

Second normal form (2NF) is based on the concept of full functional dependency.

Full Dependency: A functional dependency X -> Y is a full functional dependency if removal of any attribute A from X means that the dependency does not hold any more; that is, for any attribute A X, (X - {A}) does not functionally determine Y.

Partial Dependency: A functional dependency X -> Y is a partial dependency if some attribute A can be removed from X and the dependency still holds; that is, for some A X, (X - {A}) -> Y).

EXAMPLE: ROLLNO,COURSENO -> MARKS ….. FFD
ROLLNO,~~NAME~~-> ADDRESS……. PFD



D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

2NF

Second normal form (2NF) is based on the concept of full functional dependency. 2NF states that no non prime attribute must be partially dependent on key of table

EID	PNUMBER	HOURS	ENAME	PNAME	PLOCATION

Given FD: {EID, PNUMBER} → HOURS
EID → ENAME
PNUMBER → PNAME, PLOCATION

{SSN, PNUMBER} -> HOURS is a full dependency (neither SSN -> HOURS nor PNUMBER -> HOURS holds). However, the dependency EID → ENAME and PNUMBER → PNAME, PLOCATION is partial.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Second normal form (2NF) is based on the concept of full functional dependency. 2NF states that no non prime attribute must be partially dependent on key of table

EID	PNUMBER	HOURS

{EID, PNUMBER} → HOURS

EID	ENAME

EID → ENAME

PNUMBER	PNAME	PLOCATION

PNUMBER → PNAME, PLOCATION

Now all three tables are in 2NF

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

EXAMPLE 2NF

Patient_Drug				
Pno	Drugno	No_of_units	Pname	DrugName
P1	d1	10	Krishna	Dollo
P1	d2	20	Krishna	Paracetomol
P1	d3	15	Krishna	Crocin
P1	d4	25	Krishna	Nefdyl
P2	d3	30	Rama	Crocin
P2	d4	20	Rama	Nefdyl

HERE LOT OF REDUNDANCIES ARE PRESENT: KRISHNA, d3 etc.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

FD's = {
Pno, Drugno → No_of_units .....FFD
Pno → Pname .....PFD
Drugno → DrugName .....PFD because as follows,
}

- Prime attributes keys: { Pno, Drugno}+= {Pno, Drugno, No_of_units, Pname, Drugname}
- We can say that from Pno & Drugno we can find any of the attributes from the table
- ➢ Create tables: **Relation 1** = {Pno}+= {Pno, Pname}
- **Relation 2**= {Drugno}+={DrugName}
- So original table **Patient_Drug** will have attributes as follows: attributes not present in the Relation 1 & 2 tables.

Relation 3 = {Pno,Druno}-> {Pno,Drugno,No_of_units}

D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Relation 1,2 ,3 are as follows:

Patient

Pno	Pname
P1	Krishna
P2	Rama

Drugs

Drugno	DrugName
d1	Dollo
d2	Paracetomol
d3	Crocin
d4	Nefdyl

Patient_Drug

Pno	Drugno	No_of_units
P1	d1	10
P1	d2	20
P1	d3	15
P1	d4	25
P2	d3	30
P2	d4	20

Note: to avoid the redundancy in data we divide the table to satisfy the 2NF conditions. Here the repletion of the data is reduced. Also update anomalies also avoided like if you want to update the name of patient from Krishna to Ramesh then you have to update in on table only.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

2NF

- **Convert it into 2NF form**

```
R ( ABCDE)
FD's = {     AB → C ,
             A  → D
             B  → E
        }
```

- **Find the key: means attributes not present in the RHS side.**

{A,B}+={A,B,C,D,E}

FROM above given FDS & and prime attributes keys we can say that

```
AB → C ,
A  → D
B  → E
```

…….. FFD

……..PFD

……..PFD

Because non prime attribute could be derived from partial prime key attributes

D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

Also find {A}+={A,D} & {B}+={B,E}

- SO,the relations possible are as follows:

Relation 1= {A}+={A,D}

Relation 2= {B}+={B,E}

Relation 3= {A,B}+= {A,B,C}

NOTE: in relation3 only those attributes are included which are not in the relation 1 & 2 like 'C' & with the prime key attributes 'AB'.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Example 2NF

R (ABCDE)
FD's = { BC → A,
B → D
D → E
}

- PRIME ATTRIBUTE KEY: {BC}+={B,C,A,D,E}

SO, BC is the prime attribute key because using BC its possible to find any of the attributes.

- Find full & partial functional dependencies in given FD.

from prime keys we can say that BC-> A is FFD, B->D is PFD, D->E is D is not prime key attribute so its default to FFD.

- Find the closure for {B}+

{B}+= {B,D,E}

- Find relations:

relation1: {B}+= {B,D,E} and **relation2**: {BC}+={B,C,A}

- **NOW, RELATION 1 & 2 BOTH ARE IN THE 2NF FORM**

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Example 2NF

R (ABCDE)
FD's = { A → B ,
C → D
}

- PRIME ATTRIBUTE KEY: {ACE}+={A,C,E,B,D}

SO, ACE is the prime attribute key because using ACE its possible to find any of the attributes.

- Find full & partial functional dependencies in given FD.

from prime attribute keys we can say that A-> B is PFD, C->D is PFD

- Find the closure for {A}+ & {C}+

{A}+= {A,B} & {C}+= {C,D}

- Find relations:

relation1: {A}+= {A,B} and **relation2**: {C}+={C,D} and realtion3: {A,C,E}+=(A,C,E)

- **NOW, RELATION 1 & 2 BOTH ARE IN THE 2NF FORM**

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Lecture No: 30: 3NF & BCNF NORMALIZATION



3NF

Third normal form (3NF) is based on the concept of transitive dependency. A functional dependency X -> Y in a relation schema R is a transitive dependency if there is a set of attributes Z that is neither a candidate key nor a subset of any key of R and both X -> Z and Z -> Y hold.

Definition : A Relation Schema is said to be in 3NF

a) It is in 2 NF

b) No non prime attribute should be transitively dependent on Candidate Key

Let R(X, Y, Z) be a table and having FD = { X -> Y, Y -> Z }, determine whether the given R is in 3NF or not ? If not then decompose it into 3NF

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

3NF

Patient_Ward

Pno	Pname	Ward_no	Ward_Name
P1	Krishna	W1	Special
P2	Hari	W1	Special
P3	Ramu	W1	Special
P4	Kamal	W2	ICU

FD's = { Pno → Pname, Ward_no
Ward_no → Ward_Name
}

Repetition of data in table

…… Non key attribute deriving the non key attribute is transitive dependency
i.e. Ward_no deriving Ward_Name

D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

- The **prime attribute** is {Pno}+={Pno,Pname,Ward_no,Ward_Name}
- Find out the **closure** for {ward_no}+={ward_no, Warn_Name}
- **Create relations** as follows: relation 1= {ward_no}+={ward_no, Warn_Name}
  relation 2= {Pno}+={Pno,Pname,Ward_no}

PRIMARY KEY

FOREIGN KEY

Ward	
Ward_no	**Ward_Name**
W1	Special
W2	ICU

Patient_Ward		
Pno	**Pname**	**Ward_no**
P1	Krishna	W1
P2	Hari	W1
P3	Ramu	W1
P4	Kamal	W2

NOTE: ward is parent table Ward_no is primary key for it. In Patient_Ward child table the same Ward_no will become the foreign key.

3NF are free from insert, update , delete anomalies. We can do anything in tables. That will cause issues in any other table.

D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

- 3NF always ensures the function dependencies preserving and lossless joins.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

- R(A,B,C,D)
- FUNCTIONAL DEPENDANCIES (A-> BC,
  C-> D)

SOLVED:

1. FIND THE TRANSITIVE DEPENDANCY IS PRESENT OR NOT ?

YES ITS PRESENT BCZ A DERIVES BC & C DERIVES D. SO ITS NOT IN 3NF FORM.

SO CHECK FOR 2NF FORM OR NOT BY CHECKING THE PARTIAL & FULL DEPENDENCIES.

A-> BC……. FFD

C->D……..FFD

SO GIVEN FDS ARE IN 2NF FORM.

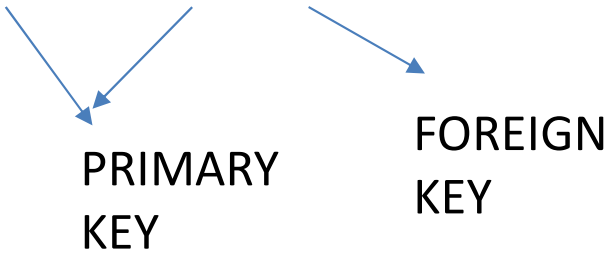
D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

2. NOW CONVERT 2NF INTO 3NF. WE HAVE ISSUE RELATED TO TRANSITIVE DEOENDENCY SO FIND PRIME ATTRIBUTE KEY & THEN CLOSURE FOR TRANSITIVE DEPENDENCY PROBLEM FD. FROM R.H.S. OF THE GIVEN FDS , THE A IS PRIME KEY ATTRIBUTE. SO CLOSURE AS FOLLOWS

- SO A+={A,B,C} & C+={C,D}

3. START WRITING RELATIONS:

R1(C,D) & R(A,B,C)



D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

- R(A,B,C,D,E)
- FUNCTIONAL DEPENDANCIES (A-> B,

C->D

B-> E)

SOLUTION:

1. FIND WHETHER IN 3NF ?

SEE FOR TRANSITIVE DEPENDANCY. Yes transitive dependency is present.

A->B & B-> E…… i.e. its not in 3NF form

Now check for whether given fds are in 2NF or not?

So check for partial & full dependencies

Find the prime key attribute= AC. So, AC+={A,C,B,E,D}

So A->B ……... Partial FD

C-> D …… partial FD

B->E…….. Full FD

Here the partial dependencies are present so its not in 2NF form.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

2. **FIND CLOSURE FOR THE PROBLEM OF PARTIAL FDS**

- To get it 2NF from find the closure for the partial FDs
- A->B= A+={A,B,E}
- C->D= C+={C,D}

3. CREATE RELATIONS/tables AS FOLLOWS

R1(A,B,E) & R2(C,D) & R(A,C)

NOW ABOVE RELATION IS IN THE 2NF FORM.

But we have to convert it into 3NF form so check for the transitive dependencies present in the relation R1 & R2.

Yes T.D. is present in R1 as A-> B & B-> E.

So find out the closure for the transitive dependencies in relation R1.

We can write/ split it as follows:

R1=(A,B) & R1'=(B,E)

4. **Finally we have tables as follows**: R1=(A,B) & R1'=(B,E) & R(A,C) & R2(C,D)

D Y PATIL RAMRAO ADIK INSTITUTE OF TECHNOLOGY NAVI MUMBAI

General steps are as follows for Normalization

- Find the functional dependencies from table.
- Find out the prime attribute key to know the full and partial functional dependencies to identify the 2NF. Also find transitive dependencies to identify the 3NF form.
- After identifying the current form of the relation/ FDs try to find the closure for the wrong FDs. And then form relations from the closures. One main original relation will have the all attributes remaining from the previously generated relations.
- Now again check for the form converted with 1 step up. If still some wrong FDs are present then directly find the closure and form relations. Finally after reaching to 3NF the all the relation will be linked with each other with primary OR foreign key.

D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Boyce codd normal form(BCNF)

- A relation is in BCNF the following condition hold in every non-trivial function dependency X->Y where X is super key & it must be in the 3NF form.



D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Thank You



D Y PATIL
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI